

1. Which of the following statements is FALSE:

SNP refers to a Single Non-defined Polymorphism

SNV refers to a Single Nucleotide Variant.

Structural variants include block deletions and insertions, among others.

SNVs encompass single nucleotide insertions, deletions and substitutions.

2. Which of the following statements is FALSE:

The VCF INFO lines describe characteristics of the variant, included in column 8.

The VCF format shows the changes in amino acid resulting from the nucleotide mutation, in column 3. column 3 of is ID; contains identifiers for the variant

The BCF format is a binary compressed version of VCF.

VCF stands for Variant Call Format.

3. What program can be used to generate a list of candidate sites of variation in an exome data set:

samtools mpileup

samtools tview

bwamem

head

4. In a comprehensive effort to study genome variation in a patient cohort, you sequence and call variants in the exome, whole genome shotgun and RNA-seq data from each patient. Which of the following is FALSE when comparing these three types of resources:

Exome sequencing comprehensively captures variants in the 3' and 5' UTRs of genes.

Exome sequencing can capture variants in a pre-defined set of coding exons and their immediate surrounding area.

Exome sequencing cannot determine variants in novel polymorphic alternative splicing events.

Exome sequencing captures fewer variants than whole genome sequencing.

5. Which of the following options can be used to allow bowtie2 to generate partial alignments?

--local

--sensitive

-D

-ignore-quals

6. Select the correct interpretation for the snippet of 'mpileup' output below.

```
Chr3    11700316    C    8    .$..... 8C@C;CB3
Chr3    11951491    G    16   AAAA,.....aA..A    C2@2BCBCCCAC2CC4
```

Only site 2 shows potential variation; the alternate letter for site 2 is A; site 1 has 8 supporting reads, and site 2 has 16

Only site 2 shows potential variation; the alternate letter for site 2 is A; the alternate allele for site 2 is supported by 9 reads

Only site 2 shows potential variation; the alternate letter for site 2 is G; site 1 has 8 supporting reads, and site 2 has 16

Only site 2 shows potential variation; the alternate letter for site 2 is '.'; site 1 has 8 supporting reads, and site 2 has 16

7. Given the set of variants described in the VCF excerpt below, which of the following is FALSE?

```
##INFO=<ID=DP,Number=1,Type=Integer,Description="Raw read depth">
##INFO=<ID=MQ,Number=1,Type=Integer,Description="Average mapping quality">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=PL,Number=G,Type=Integer,Description="List of Phred-
scaled genotype likelihoods">
Chr3    11966312    .    G    A    15.9    .    DP=5;MQ=15    GT:PL    1/1:43,9,0
Chr3    11972108    .    TAAAA    TAAA    32.8    .    INDEL;IDV=7;IMF=0.636364;DP=
11;MQ=22    GT:PL    0/1:66,0,2
Chr3    13792328    rs145271872    G    T    5.5    .    DP=1;MQ=40    GT:PL    0/1:32,3,0
```

The sample contains only the alternate allele for variant 3

Average mapping quality for variant 3 is 40

The sample contains only the alternate allele for variant 1

The sample contains both alleles for variant 2

8. What does the following code do:

```
bowtie2 -x species/species -U in.fastq | grep -v "^@" | cut -f3 | sort | uniq -c
```

Run bowtie2 with a set of single-end reads, reporting the top 5 alignments for a read; then list the number of matches containing insertions and deletions, respectively

Run bowtie2 with a set of single-end reads, reporting the best alignment only; then determine the number of matches on each genomic sequence

Run bowtie2 with a set of single-end reads, allowing for local matches; then determine the number of matches with unmapped mates

Run bowtie2 with a set of single-end reads, reporting up to 5 alignments per read; then determine the number of matches with unmapped mates

9. What does the following snippet of code do NOT do:

```
samtools mpileup -O -f genome.fa in.bam | cut -f7
```

Require a sorted BAM file

Produce a 7-column intermediate mpileup file that is piped to 'cut'

Report an empty column

Report in the intermediate mpileup output the qualities of all read bases aligned at that position

10. What does the following code do NOT do:

```
bcftools call -v -c -O z -o out.vcf.gz in.vcf.gz
```

Take input from a VCF compressed file

Take input from the file in.vcf.gz

Write the output to file out.vcf.gz

Report all candidate sites